

Service Configuration File

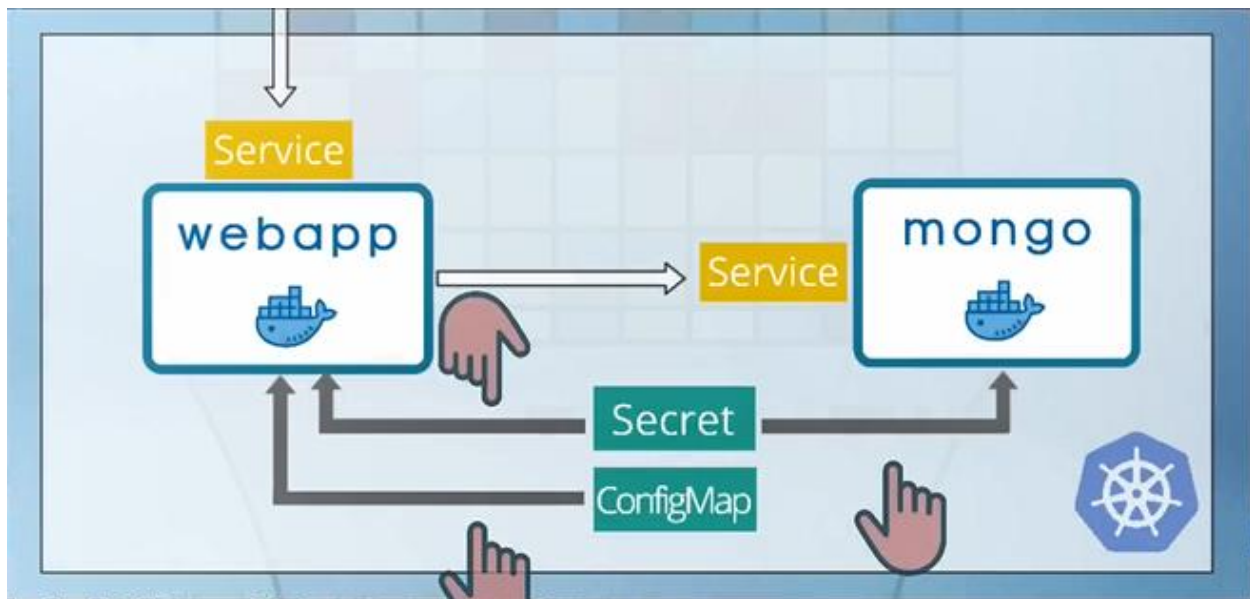
- **kind:** "Service"
- **name:** an arbitrary name
- **selector:** select pods to **forward** the requests to

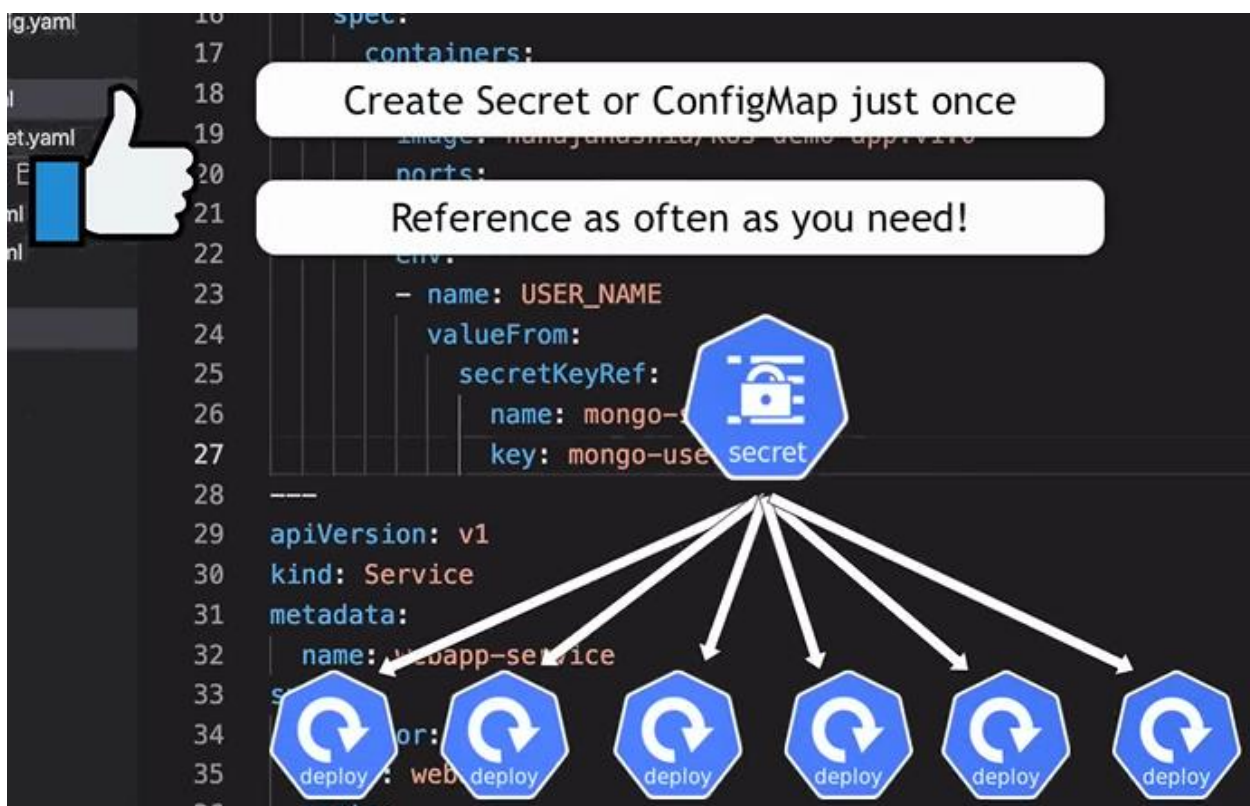
mongo-service

Service



Connecting Service to Pods





To configure ConfigMap in kubernetes:

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: mongo-config
data:
  mongo-url: mongo-service # where mongo-service is the name of service that will
  mongodb use it
```

To configure Secret in Kubernetes:

```
apiVersion: v1
kind: Secret
metadata:
  name: mongo-secret
data:
  mongo-user: bW9uZ291c2Vy
  mongo-password: bW9uZ29wYXNzd29yZA==
```

To configure db deployment with service:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo-deployment
  # For deployment labels are optional
  labels:
    app: mongo
spec:
  replicas: 1
  selector:
    # All pods that have the same label are belong to this deployment
    matchLabels:
      app: mongo
  # this is pod template the has label: mongo
  template:
    metadata:
      # for pod the labels are required
      labels:
        app: mongo
    spec:
      containers:
        # this will create pods with mongo:latest image
        - name: mongo
          image: mongo:latest
          ports:
            - containerPort: 27017
          resources: # this from qwen.ai
            requests:
              memory: "256Mi"
              cpu: "100m"
            limits:
              memory: "512Mi"
              cpu: "500m"
          env:
            - name: MONGO_INITDB_ROOT_USERNAME
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-user
            - name: MONGO_INITDB_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
```

```

        key: mongo-password
    volumeMounts: # this from qwen.ai
    - name: mongo-storage
      mountPath: /data/db
    volumes: # this from qwen.ai
    - name: mongo-storage
      emptyDir: {}

---
apiVersion: v1
kind: Service
metadata:
  # To remember only: this is the name of service that match the mongo-config
  name: mongo-service
spec:
  selector:
    app: mongo
  ports:
    - protocol: TCP
      port: 27017
      targetPort: 27017

```

To configure web app deployment and service:

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: webapp-deployment
  # For deployment labels are optional
  labels:
    app: webapp
spec:
  replicas: 1
  selector:
    # All pods that have the same label are belong to this deployment
    matchLabels:
      app: webapp
  # this is pod template the has label: webapp
  template:
    metadata:
      # for pod the labels are required
      labels:

```

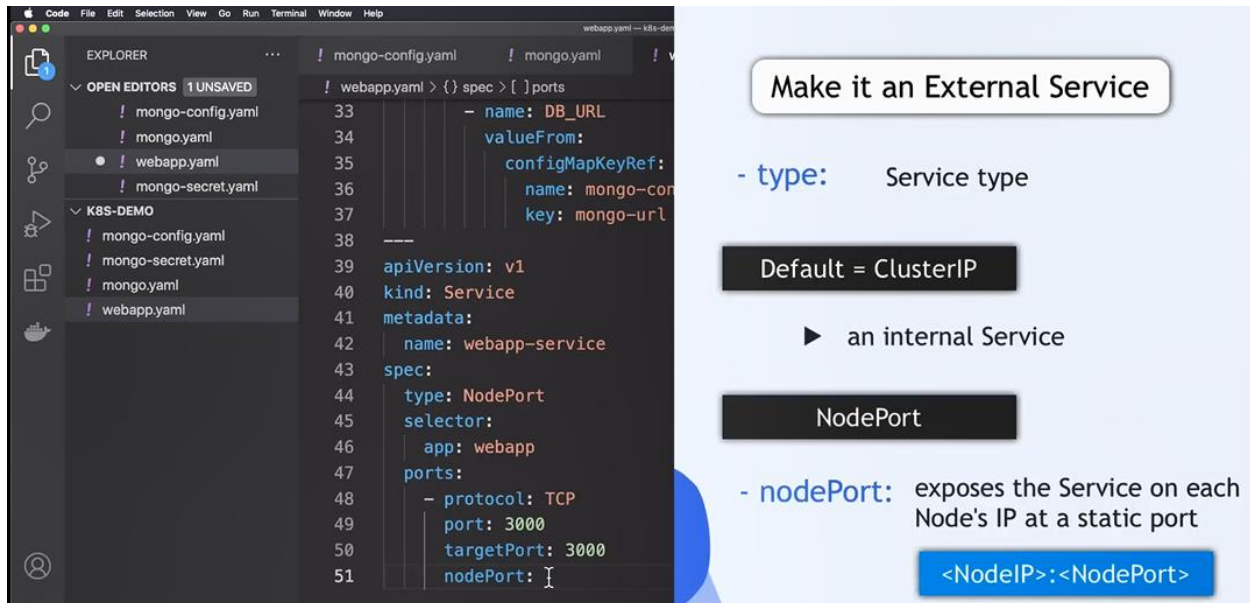
```

    app: webapp
spec:
  containers:
  - name: webapp
    image: nanajanashia/k8s-demo-app:v1.0
    ports:
    - containerPort: 3000
    resources: # this is from qwen.ai
      requests:
        memory: "256Mi"
        cpu: "100m"
      limits:
        memory: "512Mi"
        cpu: "500m"
    env:
    - name: USERNAME
      valueFrom:
        # In this way we ref from secret
        secretKeyRef:
          name: mongo-secret
          key: mongo-user
    - name: USER_PWD
      valueFrom:
        secretKeyRef:
          name: mongo-secret
          key: mongo-password
    - name: DB_URL
      valueFrom:
        # In this way we ref from map
        configMapKeyRef:
          name: mongo-config
          key: mongo-url
---
apiVersion: v1
kind: Service
metadata:
  name: webapp-service
spec:
  type: NodePort # in this way we make this service external service,
  # so we can reach to it from external host
  selector:
    app: webapp
  ports:
  - protocol: TCP
    port: 3000

```

```
targetPort: 3000
nodePort: 30100
```

NodePort: Must be between 30000 && 32767



The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'K8S-DEMO' with files: 'mongo-config.yaml', 'mongo-secret.yaml', 'mongo.yaml', and 'webapp.yaml'. The code editor shows the 'webapp.yaml' file with the following content:

```
33     - name: DB_URL
34       valueFrom:
35         configMapKeyRef:
36           name: mongo-conf
37           key: mongo-url
38   ---
39   apiVersion: v1
40   kind: Service
41   metadata:
42     name: webapp-service
43   spec:
44     type: NodePort
45     selector:
46       app: webapp
47     ports:
48     - protocol: TCP
49       port: 3000
50       targetPort: 3000
51       nodePort: I
```

To the right of the code editor is a diagram titled 'Make it an External Service'. It explains the 'type' field in the service specification:

- **type:** Service type
- Default = ClusterIP
- an internal Service
- NodePort**
- **nodePort:** exposes the Service on each Node's IP at a static port
- <NodeIP>:<NodePort>

ConfigMap and Secret must exist before Deployments

apply manages applications through files defining K8s resources

```
kubectl apply -f <file-name.yaml>
```

```
C:\Tests\Docker-devops-kubernetes\Tutorial-01-kubernetes>kubectl apply -f mongo-config.yaml
configmap/mongo-config created
```

```
C:\Tests\Docker-devops-kubernetes\Tutorial-01-kubernetes>kubectl apply -f webapp.yaml
deployment.apps/webapp-deployment created
service/webapp-service unchanged
```

```

C:\Users\Tatweer>kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/mongo-deployment-587b96c5df-5pb4r  1/1     Running   0           71s
pod/webapp-deployment-79c89c576f-z48nc  1/1     Running   0           8m3s

NAME                                TYPE               CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
service/kubernetes                  ClusterIP         10.96.0.1    <none>        443/TCP          321d
service/mongo-service               ClusterIP         10.107.77.195 <none>        27017/TCP        10m
service/webapp-service              NodePort          10.110.69.126 <none>        3000:30100/TCP   10m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/mongo-deployment    1/1     1             1           10m
deployment.apps/webapp-deployment    1/1     1             1           8m3s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/mongo-deployment-587b96c5df  1         1         1       71s
replicaset.apps/mongo-deployment-66c4fbf4cd  0         0         0       10m
replicaset.apps/webapp-deployment-79c89c576f  1         1         1       8m3s

C:\Users\Tatweer>kubectl get configmap
NAME            DATA   AGE
kube-root-ca.crt  1       321d
mongo-config     1       11m

C:\Users\Tatweer>kubectl get secret
NAME            TYPE      DATA   AGE
mongo-secret    Opaque    2       11m

```

Get detailed information with "describe" command

```
kubectl describe <resourceType> <resourceName>
```

```
[\\W]$ kubectl describe service webapp-service
Name:                webapp-service
Namespace:           default
Labels:              <none>
Annotations:         <none>
Selector:            app=webapp
Type:               NodePort
IP:                 10.110.153.3
Port:               <unset> 3000/TCP
TargetPort:         3000/TCP
NodePort:           <unset> 30100/TCP
Endpoints:          172.17.0.4:3000
Session Affinity:    None
External Traffic Policy: Cluster
Events:             <none>
[\\W]$
```

View logs of container

kubectl logs <podName>

```
C:\Users\Tatweer>kubectl get pod
NAME                                READY    STATUS    RESTARTS   AGE
mongo-deployment-587b96c5df-5pb4r  1/1     Running   0          9m2s
webapp-deployment-79c89c576f-z48nc  1/1     Running   0          15m

C:\Users\Tatweer>kubectl logs webapp-deployment-79c89c576f-z48nc
app listening on port 3000!
```

```
C:\Users\Tatweer>kubectl get svc
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes      ClusterIP   10.96.0.1     <none>         443/TCP          321d
mongo-service   ClusterIP   10.107.77.195 <none>         27017/TCP        19m
webapp-service  NodePort    10.110.69.126 <none>         3000:30100/TCP  19m
```

**NodePort Service is accessible on each
Worker Node's IP address**
